Last Time:

- Attitude Representations
- Lie Group
- Quaternion Intro

Today:

- Lie Algebra + Exponential
- Quaternions in the Plane
- Quaternion Geometry in 3D

Exponential + Lie Algebra:

- Recall Rotation Matrix Kinematics:

$$\dot{Q} = Q\,S(\omega)$$

- Linear 1st order ODE if ω is constant:

$$\underbrace{\dot{Q}}_{\dot{x}} = \underbrace{Q}_{x}\underbrace{S(\omega)}_{A} \quad \Longleftrightarrow \quad \dot{x} = Ax \implies x(t) = e^{At}x_0$$

$$\Downarrow$$

$$Q(t) = Q_0 e^{S(\omega)t}$$

matrix exponential (expm in MatlabO/Python)

- $\omega\Delta t$ is an axis-angle vector $\phi = a\theta$ ← angle in radians
  (unit vector axis)

- Exponential maps from axis-angle vectors to rotation
- Useful for e.g. sampling random rotations

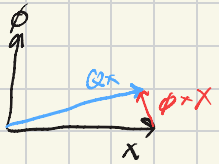
- Matrix log maps from rotation matrix to axis-angle:

$$S(\phi) = \log(Q)$$

(log in for Matlab/Python etc.)

- For small rotations $\|\phi\| << 1$:

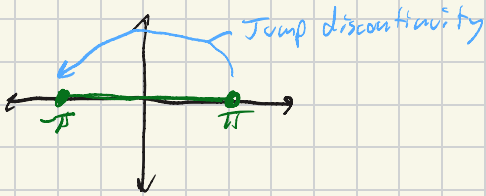
$$Q = e^{S(\phi)} \approx I + S(\phi) \Rightarrow Qx \approx (I + S(\phi))x \approx x + \phi \times x$$



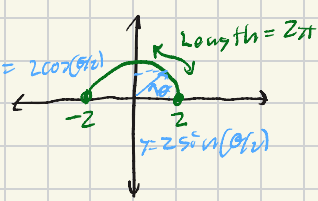
Planar Quaternions:

- Set of all possible planar rotations:

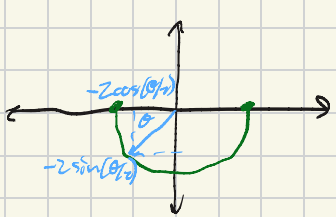
$$-\pi < \theta \leq \pi$$



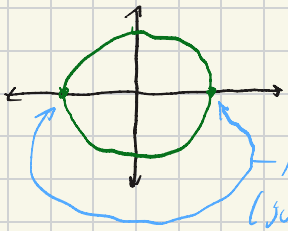
- "Pop Up"



- Copy + Reflect through origin:



- Glue copies together:



* For convenience, normalize so vectors have unit length.

* Note: $q(\theta) = \begin{bmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{bmatrix} \Rightarrow q(\theta + 2\pi) = \begin{bmatrix} \cos(\theta/2 + \pi) \\ \sin(\theta/2 + \pi) \end{bmatrix} = -q(\theta)$

$\Rightarrow$ q and $-q$ represent the same rotation

* We say that quaternions "double cover" the rotations

Combining Rotations:

$$\theta_3 = \theta_1 + \theta_2$$

$$\Rightarrow q_3 = \begin{bmatrix} \cos\left(\frac{\theta_1+\theta_2}{2}\right) \\ \sin\left(\frac{\theta_1+\theta_2}{2}\right) \end{bmatrix} = \begin{bmatrix} \cos\left(\frac{\theta_1+\theta_2}{2}\right) & -\sin\left(\frac{\theta_1+\theta_2}{2}\right) \\ \sin\left(\frac{\theta_1+\theta_2}{2}\right) & \cos\left(\frac{\theta_1+\theta_2}{2}\right) \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

- We can also recognize this as a 2D rotation matrix applied to $q(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\Rightarrow q_3 = \underbrace{\begin{bmatrix} \cos(\theta_2/2) & -\sin(\theta_2/2) \\ \sin(\theta_2/2) & \cos(\theta_2/2) \end{bmatrix}}_{L(q_2)} \underbrace{\begin{bmatrix} \cos(\theta_1/2) \\ \sin(\theta_1/2) \end{bmatrix}}_{q(\theta_1)} = \begin{bmatrix} C_2C_1 - S_2S_1 \\ S_2C_1 - C_2S_1 \end{bmatrix}$$

- This is equivalent to complex multiplication:

$$(c_1 + s_1 i) \cdot (c_2 + s_2 i) = (c_1 c_2 - s_1 s_2) + i(c_1 s_2 + c_2 s_1)$$

- We'll introduce the notation $L(q)$:

$$q_3 = L(q_2)\, q_1$$

Kinematics:

- Rotations have discontinuities/singularities, but angular velocity does not:

$\Rightarrow$ use $q \in \mathbb{R}^2$ for attitude but $\omega \in \mathbb{R}^3$ for velocity

* Integrating a gyro again:

$$\omega \rightarrow \dot{q} \rightarrow q(t)$$

$$q = \begin{bmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{bmatrix} \Rightarrow \frac{\partial q}{\partial \theta} = \begin{bmatrix} -\frac{1}{2}\sin(\theta/2) \\ \frac{1}{2}\cos(\theta/2) \end{bmatrix}$$

$$\dot{q} = \frac{\partial q}{\partial \theta}\dot{\theta} = \frac{\partial q}{\partial \theta}\omega = \begin{bmatrix} -\frac{1}{2}\sin(\theta/2) \\ \frac{1}{2}\cos(\theta/2) \end{bmatrix}\omega$$

- Same Notation:

$$\dot{q} = \underbrace{\begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}}_{L(q)} \begin{bmatrix} 0 \\ \frac{1}{2}\omega \end{bmatrix} = \overbrace{\frac{1}{2}L(q)H}^{G(q)}\omega$$

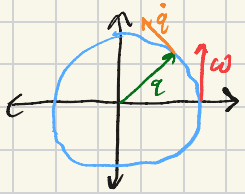
"attitude Jacobian"

$H = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Plays the same role as i (and hat map)

$\Rightarrow \quad \dot{q} = \frac{1}{2} G(q)\,\omega$

* Geometry



- q is a unit vector $\Rightarrow$ on a circle
- $\dot{q}$ is tangent to circle (orthogonal to q)
- ω is really a tangent at identity ($\theta = 0$)
- Attitude Jacobian rotates ω from tangent at $\theta = 0$ to tangent at q.

3D Rotations:

- Set of all possible axis-angle vectors $\|\phi\| \leq \pi$ is a ball of radius π
- We'll visualize this as a disc in 2D:



- Just like planar case, there's a jump when the angle gets to $\pm\pi$

- Play the same trick as before by "popping up" the disc into a hemisphere!



- Make a copy, reflect it, and glue it on to form a
- Now instead of jumping from $\pi \to -\pi$ we can continue smoothly onto the "southern hemisphere"